

C02 ASSESSMENT TOOL -2

SCENARIO BASED QUESTIONS

NAME : SHAIK SHOUIB AKTHAR

REG NO: 192312208

QUESTION:1

Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

A. Explain why this issue occurs.

B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

SOLUTION:

A. Explain why this issue occurs.

Electronic Codebook (ECB) is one of the simplest block cipher modes. In ECB mode, each block of plaintext is encrypted independently using the same secret key. This means that if the same plaintext block appears more than once, it will always produce the same ciphertext block.

In a company that transmits financial records, many data blocks such as account numbers, transaction codes, or customer details may be repeated. Since ECB encrypts identical blocks in the same way, repeated patterns remain visible in the ciphertext. An attacker can observe these repeated patterns and may identify similarities in the data, even without knowing the secret key. Therefore, ECB does not provide sufficient security for sensitive financial information.

B. Suggest a more secure block cipher mode and justify your choice.

The most suitable block cipher mode for this scenario is **Cipher Block Chaining (CBC)**.

In CBC mode, each plaintext block is first combined (XORed) with the previous ciphertext block before encryption. The first block uses an **Initialization Vector (IV)**, which is a random value used only once. Because of this chaining process, even if two plaintext blocks are identical, their ciphertext blocks will be completely different.

Advantages of CBC Mode

- Hides repeated patterns in the plaintext.
- Provides stronger confidentiality than ECB.
- Makes cryptanalysis more difficult for attackers.
- Suitable for financial transactions and confidential business data.

Conclusion: CBC mode is more secure than ECB because it prevents attackers from identifying repeated data patterns in encrypted financial information.

QUESTION 2:

Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

A. Identify the most suitable algorithm for this scenario.

B. Justify your choice based on security strength and efficiency.

SOLUTION:

A. Identify the most suitable algorithm for this scenario.

The most suitable encryption algorithm for the banking application is Blowfish.

B. Justify your choice based on security strength and efficiency.

A banking application requires strong security to protect customer information and fast encryption to process transactions efficiently.

DES (Data Encryption Standard)

- Uses a 56-bit key.
- Provides fast encryption.
- Not secure against modern brute-force attacks because the key size is too small.

3-DES (Triple DES)

- Applies the DES algorithm three times.
- Uses an effective key size of 168 bits.
- Offers much better security than DES.
- Encryption is slow because three rounds of DES are performed.

Blowfish

- Supports key sizes from 32 to 448 bits.
- Provides strong security.
- Encrypts data faster than 3-DES.
- Requires less processing time while maintaining high security.

Comparison

Algorithm	Security	Performance
DES	Low	Fast
3-DES	Very High	Slow
Blowfish	High	Fast

Conclusion: Blowfish is the best choice because it provides strong encryption with high speed, making it suitable for banking applications where both security and performance are important.

QUESTION 3:

Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

A. Explain how the key exchange works in this scenario.

B. Identify a possible security threat and suggest a solution to mitigate it.

SOLUTION:

A. Explain how the key exchange works in this scenario.

Diffie-Hellman Key Exchange allows two users to create a common secret key over a public network without sending the actual secret key.

The process is as follows:

1. Both users publicly agree on a prime number (p) and a generator (g).
2. User A selects a private key and calculates a public key.
3. User B also selects a private key and calculates a public key.
4. They exchange only their public keys through the public network.
5. User A uses User B's public key and their own private key to generate a shared secret.

6. User B uses User A's public key and their own private key to generate the same shared secret.
7. Both users now possess the same secret key, which is used to encrypt future communication.

The actual secret key is never transmitted over the network, making the communication secure.

B. Identify a possible security threat and suggest a solution.

The major security threat is the Man-in-the-Middle (MITM) attack.

In this attack, an attacker intercepts the exchanged public keys and replaces them with fake public keys. User A unknowingly shares a secret with the attacker, and User B also shares another secret with the attacker. The attacker can then decrypt, modify, and re-encrypt messages without either user knowing.

Methods to Prevent MITM Attack

- Use Digital Certificates.
- Use Digital Signatures.
- Use TLS/SSL protocols.
- Authenticate users before exchanging public keys.

Conclusion: Authentication techniques ensure that the exchanged public keys belong to the intended users, preventing attackers from intercepting the communication.

QUESTION 4:

Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

A. Describe how RSA can be used for encryption and decryption in this scenario.

B. Explain how key distribution and management play a role in maintaining security.

SOLUTION:

A. Describe how RSA can be used for encryption and decryption.

RSA is an asymmetric encryption algorithm that uses two different keys: a public key and a private key.

The sender encrypts the email using the receiver's public key. The encrypted email is then transmitted through the network. After receiving the email, the receiver decrypts it using their private key.

Since only the receiver possesses the private key, no unauthorized person can read the message. This ensures confidentiality and protects sensitive email communication.

B. Explain how key distribution and management maintain security.

The security of RSA depends on proper management of encryption keys.

- Public keys should be distributed through trusted Certificate Authorities (CAs).
- Private keys must remain secret and should never be shared.
- Digital certificates verify that a public key belongs to the correct person.
- Keys should be updated periodically to maintain security.
- Secure storage of private keys prevents unauthorized access.

Conclusion: Proper key distribution and management ensure secure communication by protecting the confidentiality and authenticity of email messages.

QUESTION 5:

Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication.

The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

A. Analyze the constraints of IoT devices.

B. Recommend a suitable cryptographic approach and justify your answer.

SOLUTION:

A. Analyze the constraints of IoT devices.

IoT (Internet of Things) devices are designed to perform specific tasks using limited hardware resources. Their major constraints include:

- Limited processing power.
- Small memory and storage capacity.
- Low battery power.
- Limited computational capability.
- Requirement for fast and energy-efficient communication.

Because of these limitations, IoT devices require lightweight cryptographic algorithms that consume fewer resources while providing strong security.

B. Recommend a suitable cryptographic approach and justify your answer.

The most suitable cryptographic approach is Elliptic Curve Cryptography (ECC).

ECC provides the same level of security as RSA but uses much smaller key sizes. Smaller keys require less memory, less processing power, and lower energy consumption. As a result, ECC performs faster and is highly suitable for low-power devices.

Comparison of RSA and ECC

Feature	RSA	ECC
Key Size	Large (1024 bits)	Small (160 bits)
Processing Speed	Slower	Faster
Memory Requirement	High	Low
Power Consumption	High	Low
Suitable for IoT	No	Yes

Conclusion: ECC is the preferred cryptographic technique for IoT devices because it offers strong security with lower memory usage, faster processing, and reduced power consumption. This makes it ideal for smart home systems, wearable devices, wireless sensors, and other resource-constrained environments.